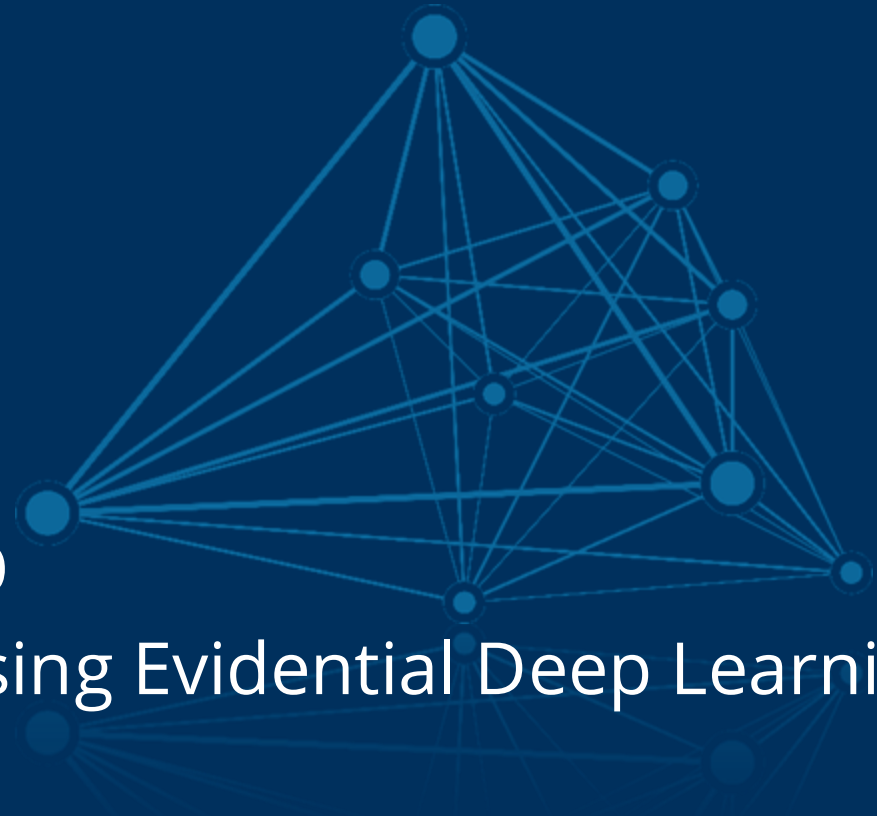


TU Dresden, Junior Professorship in Environmental Remote Sensing

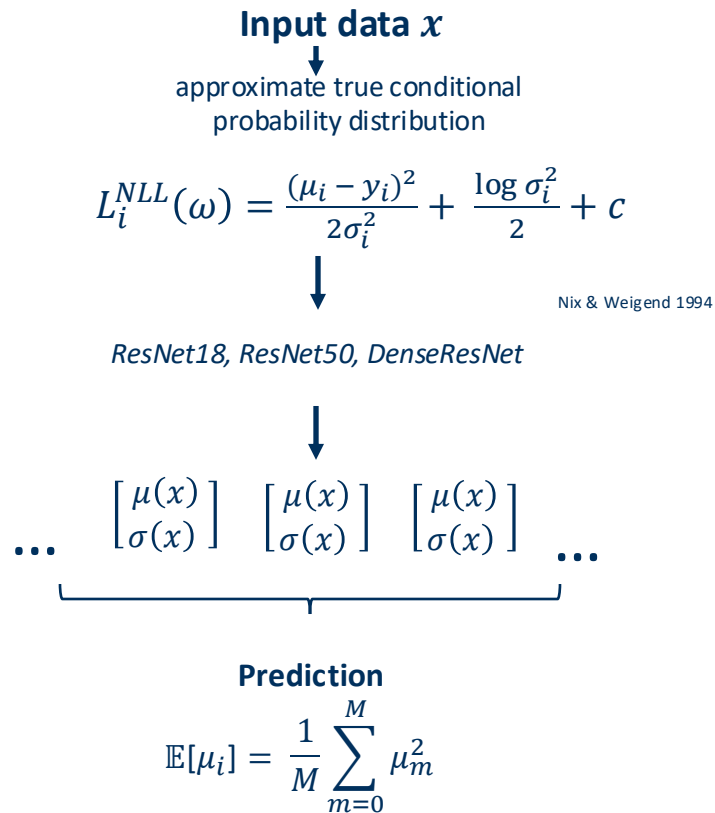
Examining the Reliability Gap

Insights into Forest Canopy Height Using Evidential Deep Learning

Lucas Kugler
Vienna EGU, 22.04.2024



Ensemble



Aleatoric uncertainty

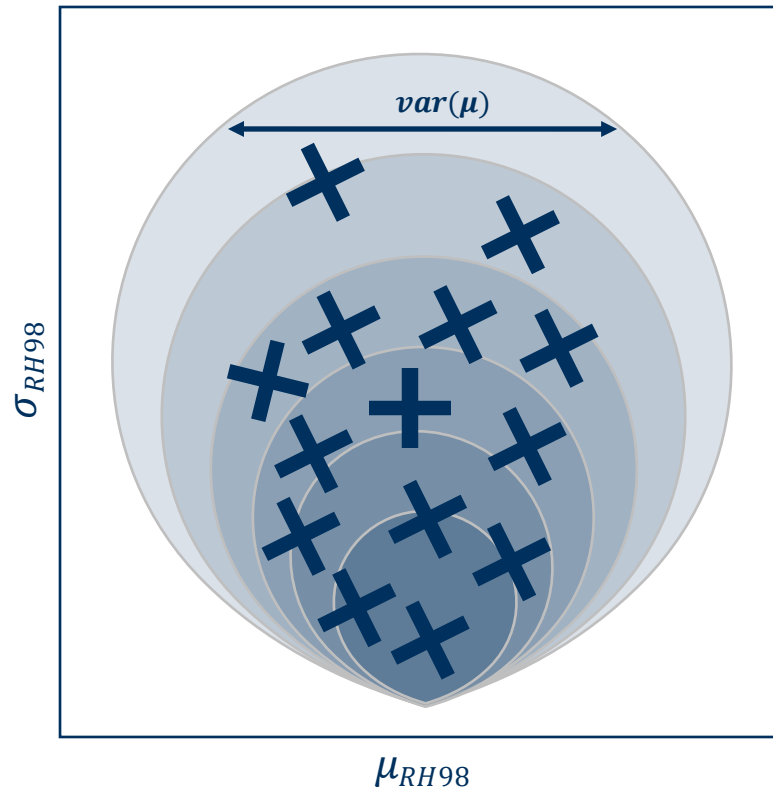
$$u_{al} = \frac{1}{M} \sum_{m=0}^M \sigma_m^2$$

Epistemic uncertainty

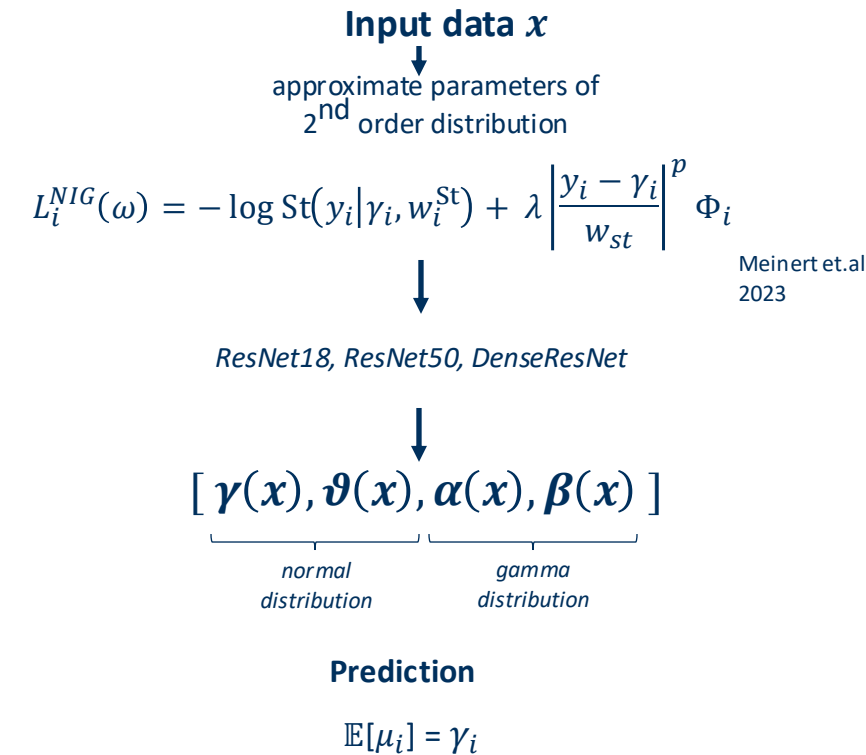
$$u_{ep} = var(\mu)$$

Lakshminarayanan 2017

$$Var[Y] = \underbrace{\mathbb{E}[\sigma^2]}_{aleatoric} + \underbrace{Var[\mu]}_{epistemic}$$



Deep Evidential Regression



Aleatoric uncertainty

$$u_{al} = \frac{\beta_i}{(\alpha_i - 1)}$$

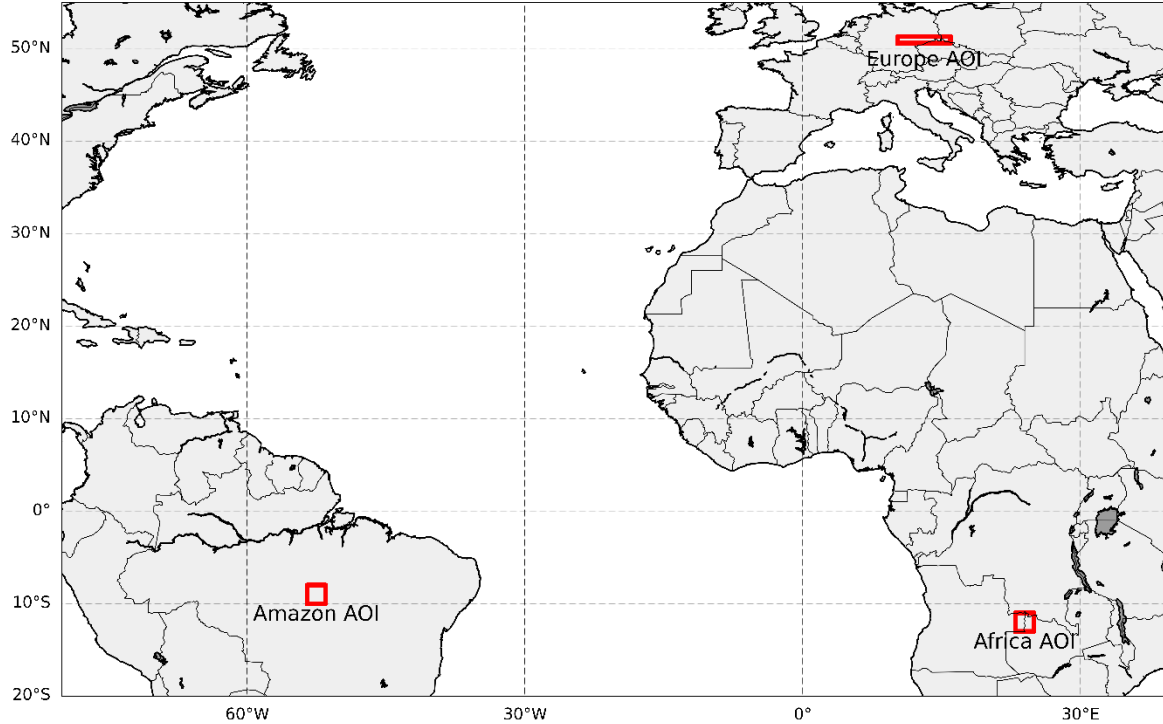
Epistemic uncertainty

$$u_{ep} = \frac{\beta_i}{\vartheta_i (\alpha_i - 1)}$$

Amini 2020 et.al

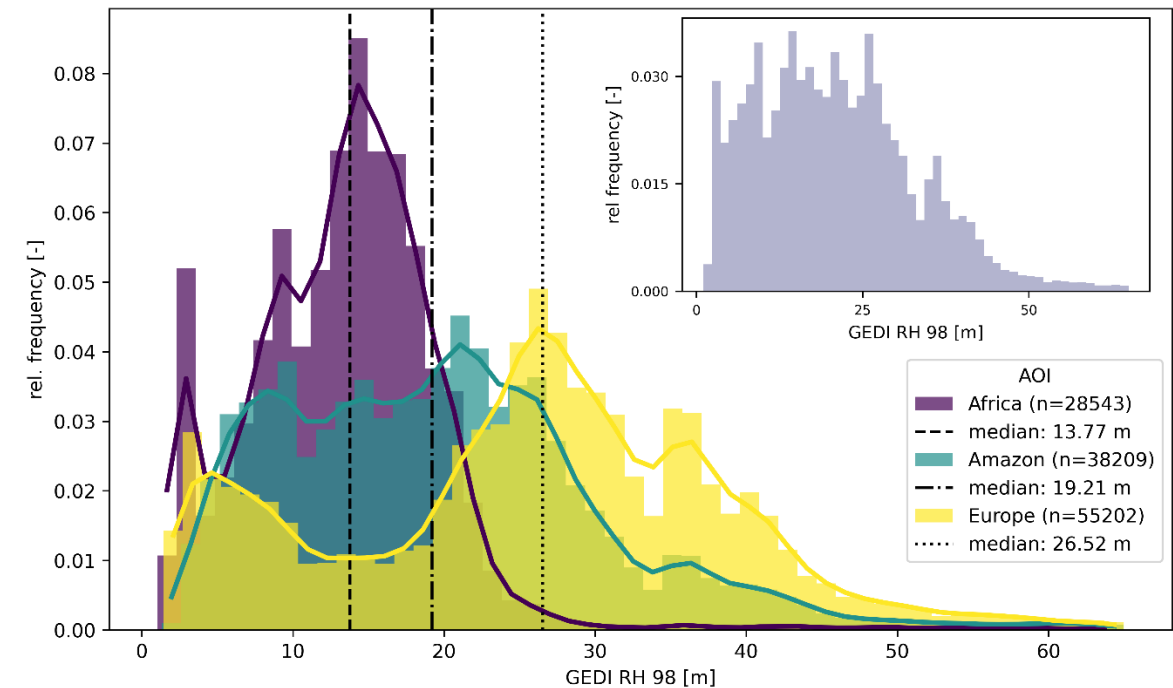
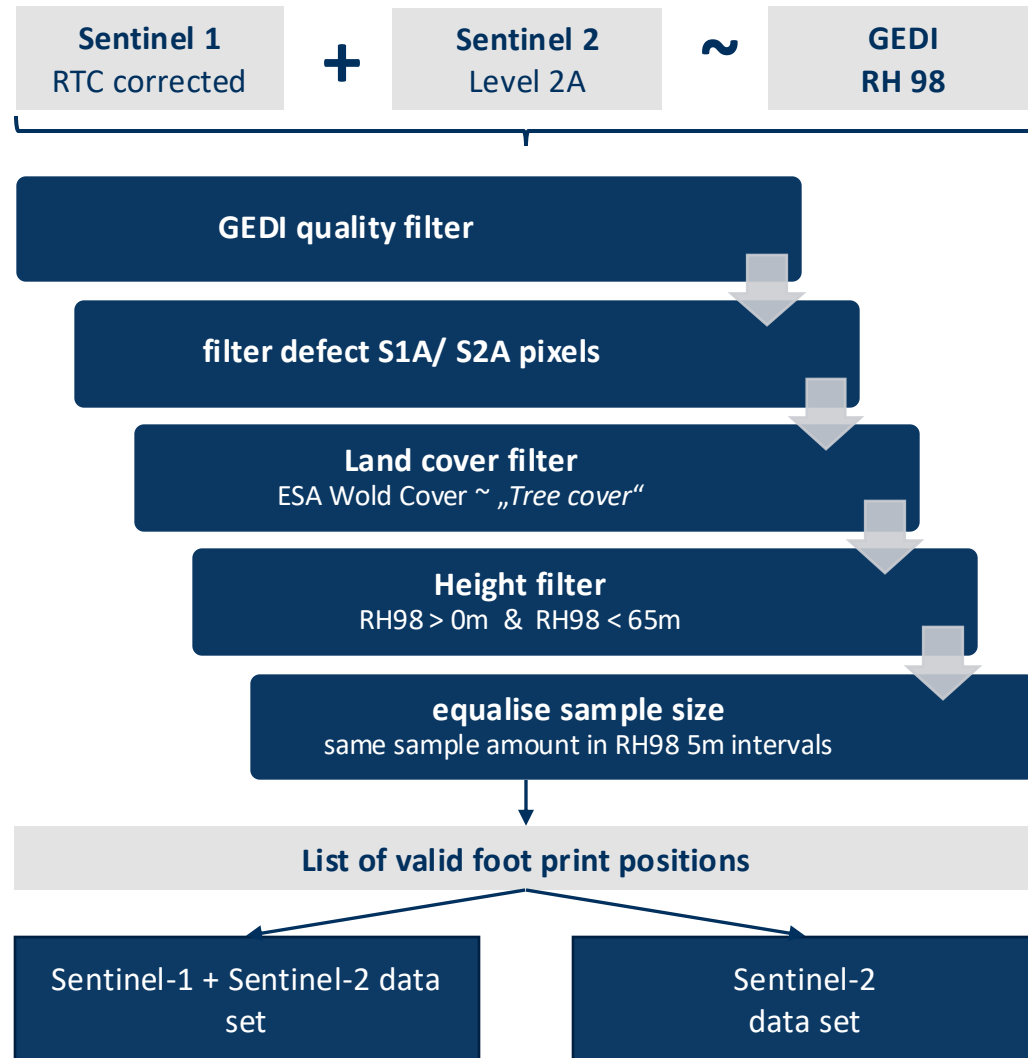
Study areas

Study areas (EPSG:4328)



	Africa 48,482 km ²	Amazon 48,955 km ²	Europe 32,827 km ²
Sentinel-2 Scenes	11	9	22
Sentinel-2 time span	15.05-15.06.21	15.05-15.06.21	15.05-15.06.21
Sentinel-1 Scenes	4	4	6
Sentinel-1 time span	15.05-15.06.21	15.05-15.06.21	15.05-15.06.21
GEDI RH 98 footprints	814,570	827,153	1,637,630
GEDI RH 98 time span	2021	2021	2021
ESA World View	2021	2021	2021

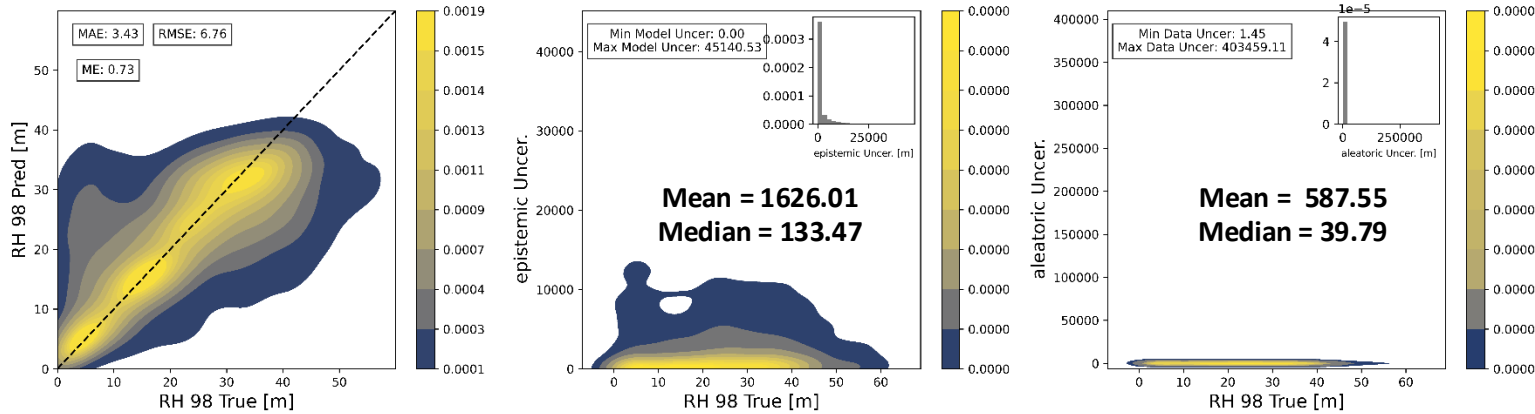
Data & Filtering



Build
5 k-folds

Redefining Uncertainties

Deep Evidential Regression (original)



Aleatoric uncertainty

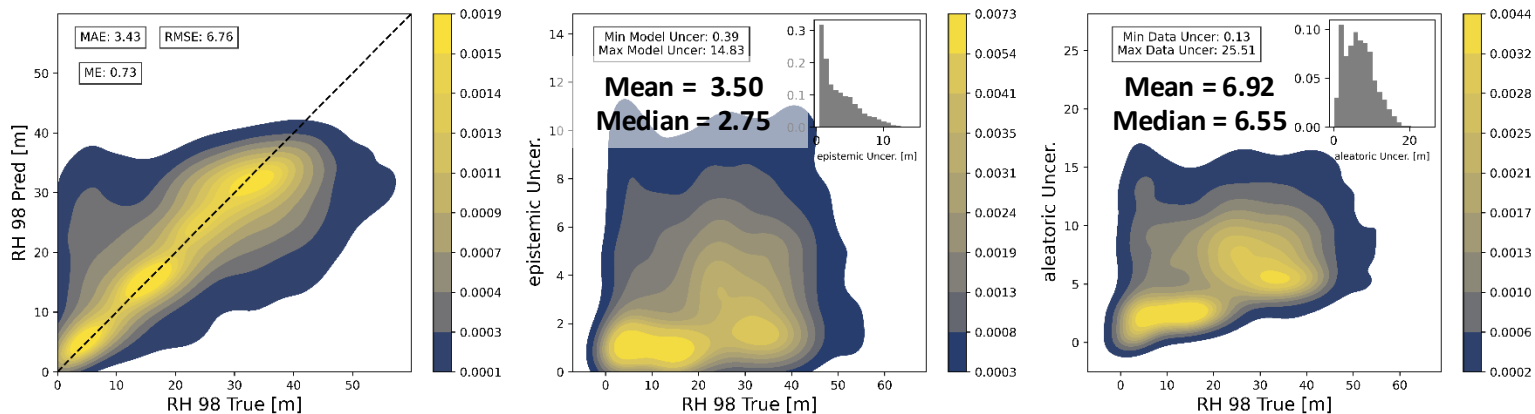
$$u_{al} = \frac{\beta_i}{(\alpha_i - 1)}$$

Epistemic uncertainty

$$u_{ep} = \frac{\beta_i}{\vartheta_i (\alpha_i - 1)}$$

Amini 2020 et.al

Deep Evidential Regression (re-definition)



Aleatoric uncertainty

$$u_{al} \equiv w_{st} = \underbrace{\sqrt{\frac{\beta_i (1 + \vartheta_i)}{\alpha_i \vartheta_i}}}_{\text{Width of the student distribution}}$$

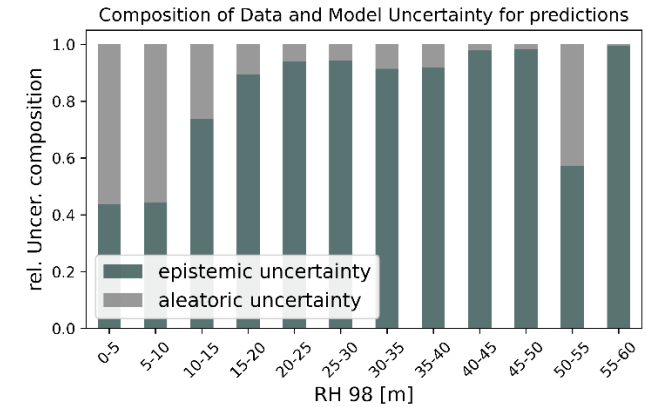
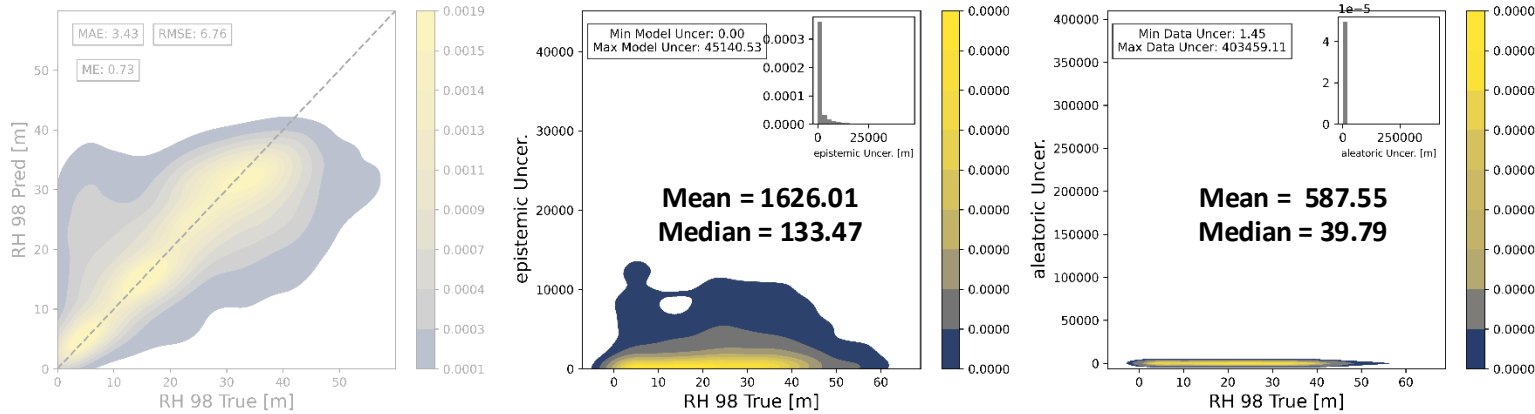
Epistemic uncertainty

$$u_{ep} = \frac{1}{\sqrt{\vartheta_i}}$$

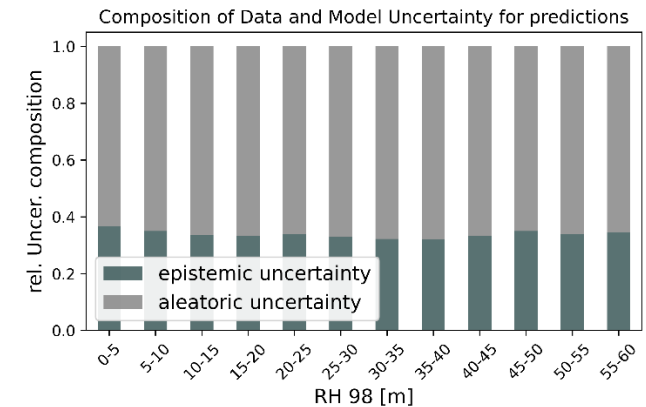
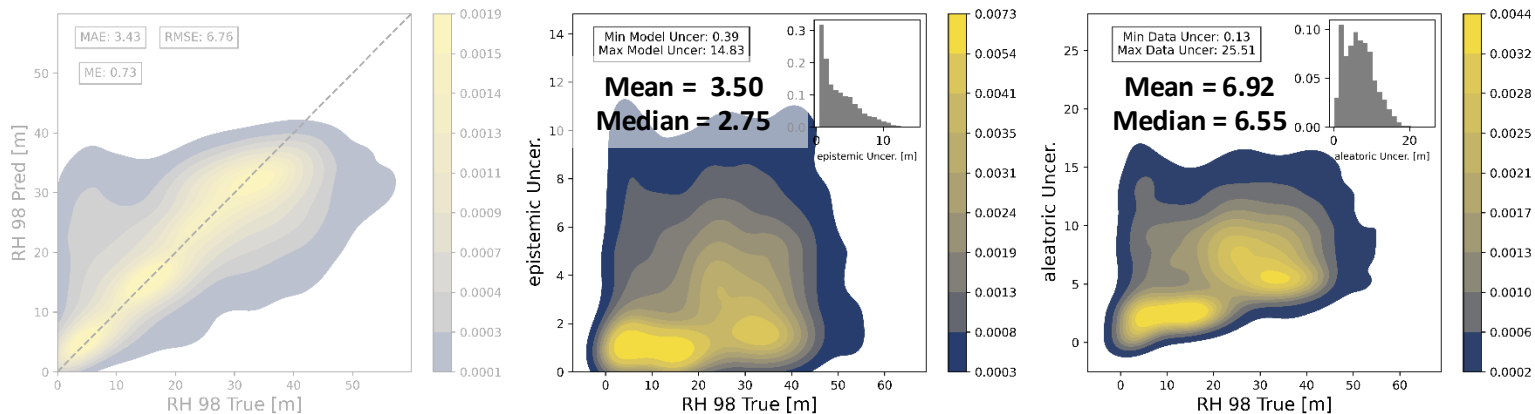
Meinert 2023 et.al

Redefining Uncertainties

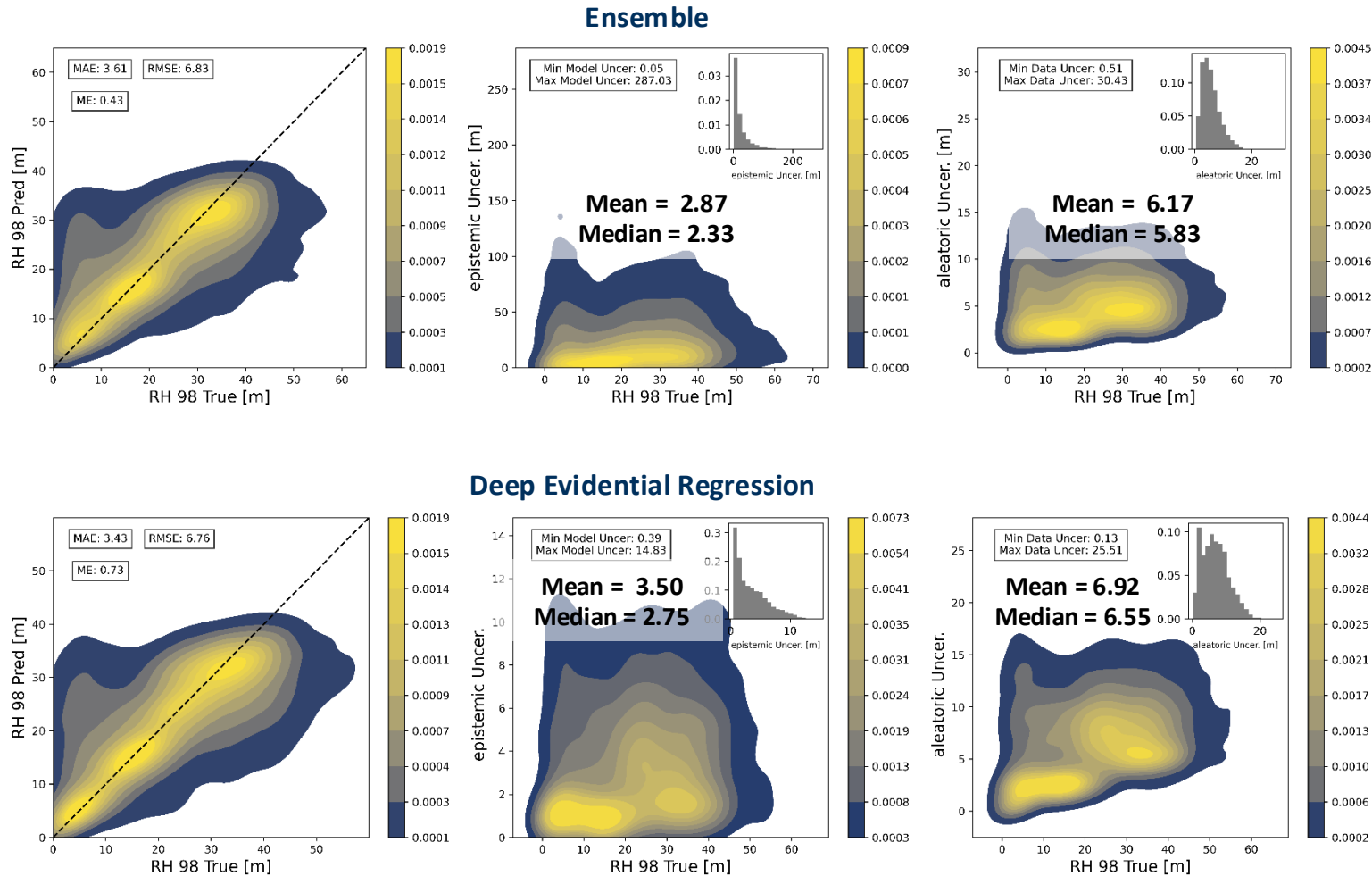
Deep Evidential Regression (original)



Deep Evidential Regression (re-definition)

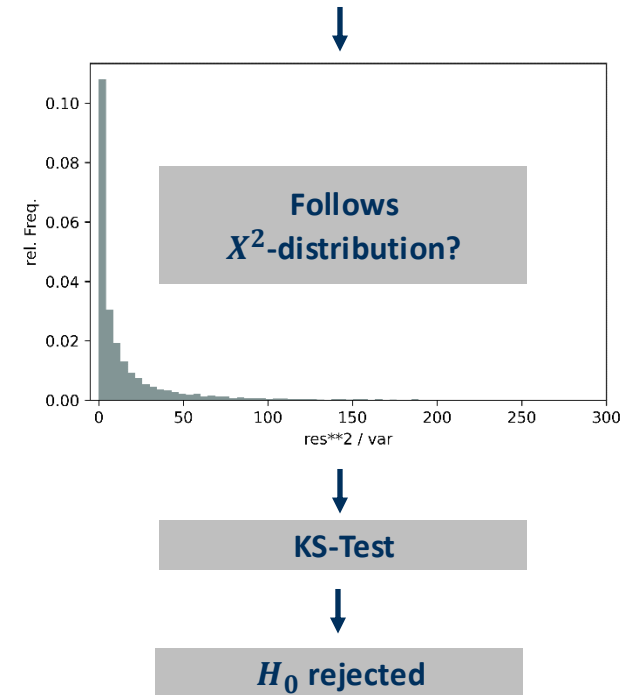


Result: Ensemble vs. DER

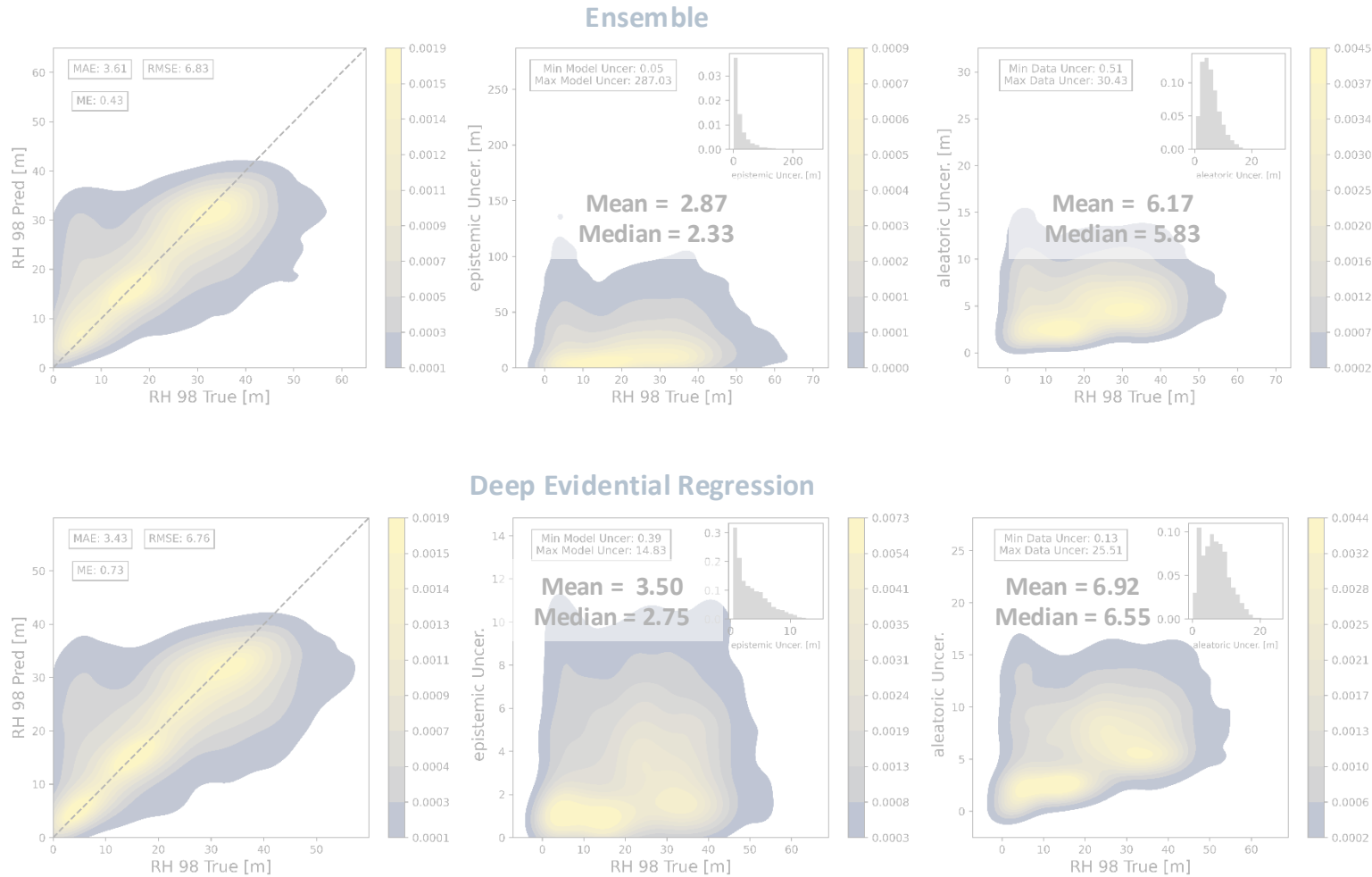


Overestimation of uncertainty?

$$\frac{(y_{pred} - y_{true})^2}{u}$$

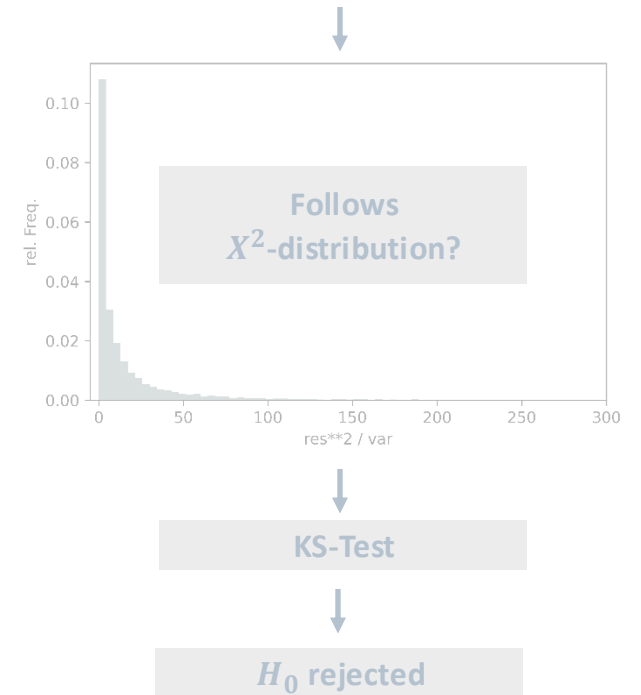


Result: Ensemble vs. DER



Overestimation of uncertainty?

$$\frac{(y_{pred} - y_{true})^2}{u}$$



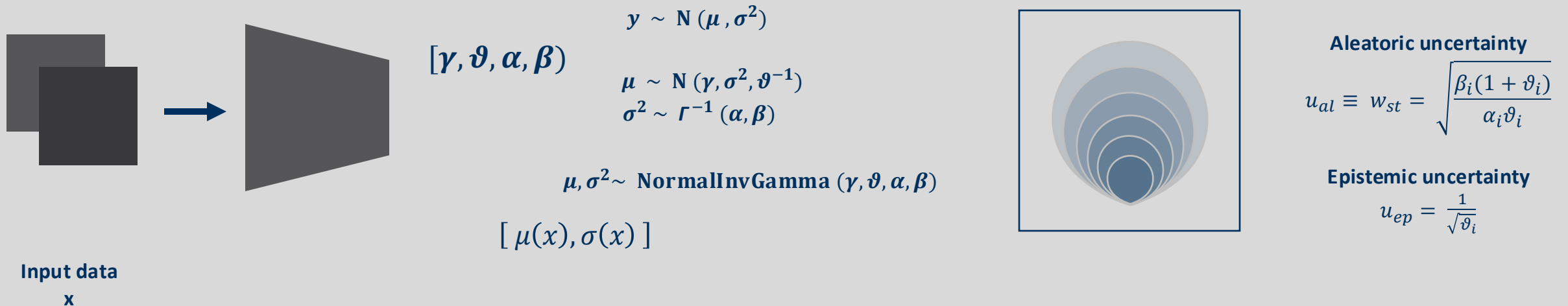
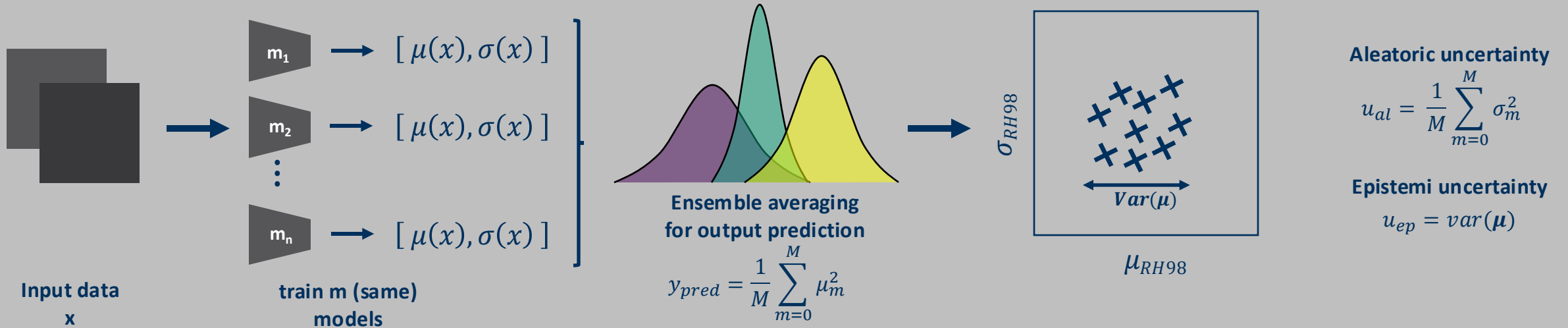


*“Knowledge is an unending adventure
at the edge of uncertainty.”*

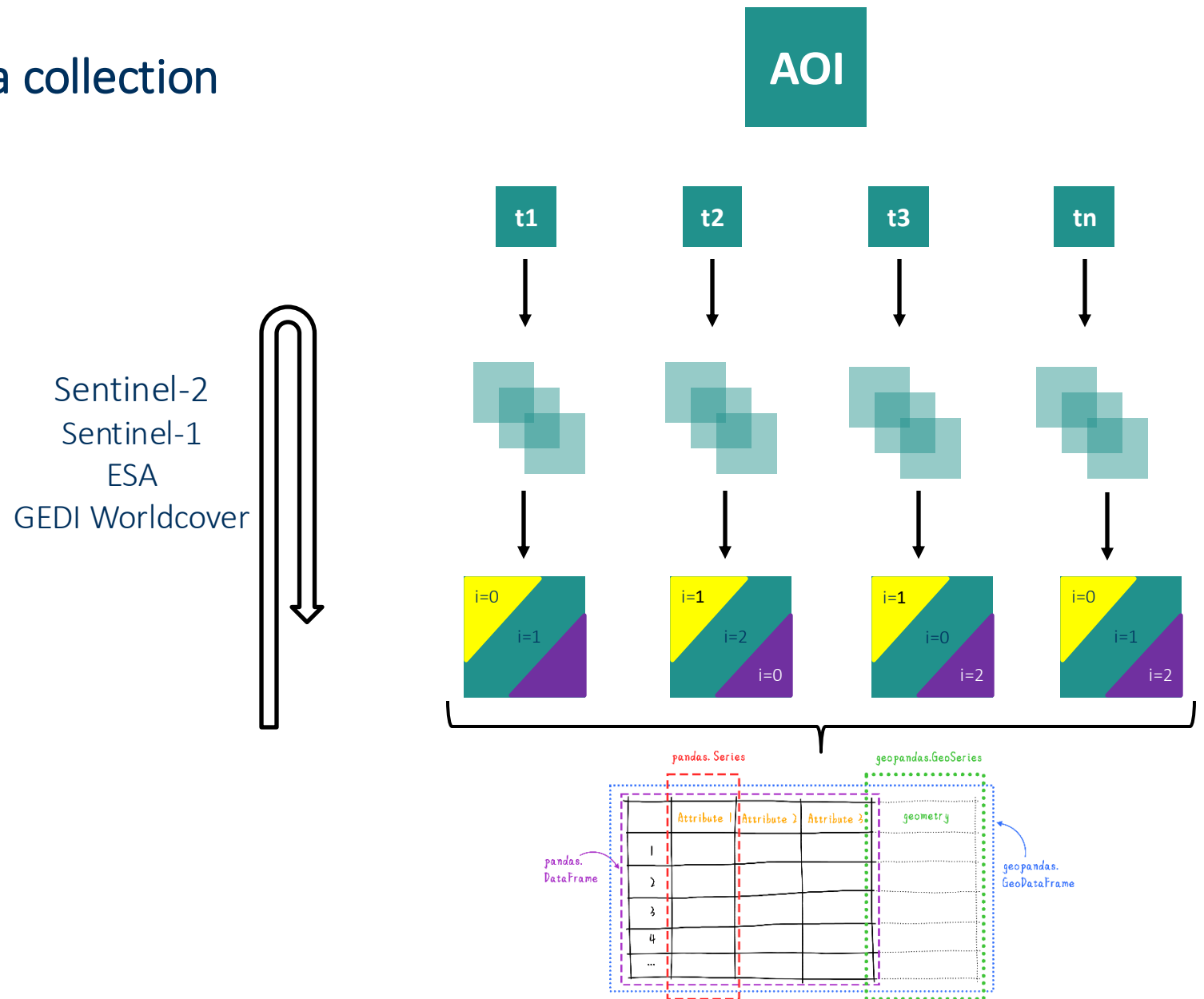
Jacob Bronowski

Concept: EDL vs. Ensemble

Ensemble to capture uncertainty



Data collection



Search items

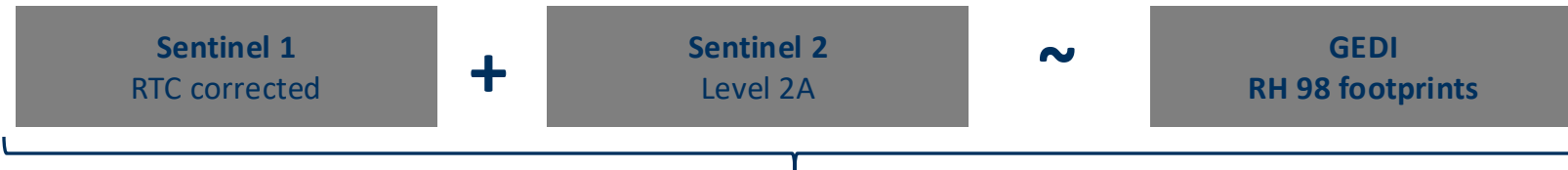
- time span
- extent
- Filter (clouds, orbit)

Items from different scenes in the time span in tile

Filter out defect and cloudy pixels and fill missing parts in first item with values from other items, filter GEDI quality

Combine S2, S1, LC and GEDI data together – build dataframe with pixel values where GEDI foot prints exists

Data & Filtering



GEDI quality
Filter for quality flag ~ 1

Land cover
ESA World cover $\sim$ Tree cover

Height filter
RH98 $> 0\text{m}$ & RH98 $< 65\text{m}$

